

CS342 Project 3 - Above and Beyond Report

Name: Joonho Ma

Extension: Keep track of wins and losses for the usernames that persists even if the server and client are closed and reopened.

As an extra component, I implemented a scoreboard, which each client can access through "Scoreboard" button on its lobby scene. The scoreboard, which is implemented through scoreboard.txt file, shows all the accumulated records(win, lose, draw) of each user. A user is listed on the scoreboard only when the user has a 'record', which means, newly connected user is not listed on the scoreboard before they complete a game. The rank is sorted by percentage of win. In case two or more users have the same win percentage as well as same number of wins, then the username is sorted in reverse alphabetical order.

Regarding the user experience, the scoreboard makes the checkers game more engaging because users can build their records over time. Instead of each game being independent, users can track their long-term performance. This can motivate users to improve their ranking and encourage them to keep playing while competing with other users.

For a potential future improvement, the current system still allows a user to enter another existing username with a high ranking if that username is not currently connected. To make the game more complete and safe, a password system could be added so that only the original user can access their own username and record.